

Technical Indicators & Strategy Backtesting with Alpaca

Ticker analyzed: SPY

Homework 2 Final Report

Ticker: SPY

Strategy descriptions

Trend Following: MACD, ADX, and moving average confirmation for directional trades.

Mean Reversion: RSI and Bollinger Bands to buy oversold pullbacks and exit rebounds.

Custom Strategy: EMA vs. SMA trend filter plus CMF, OBV slope, and RSI confirmation.

Entry and exit rules

Trend Following enters when MACD is above signal, SMA50 is above SMA200, and ADX exceeds 25.

Trend Following exits when MACD falls below signal or SMA50 falls below SMA200.

Mean Reversion enters when RSI is below 30 and price is below the lower Bollinger Band.

Mean Reversion exits when RSI is above 70 or price rises above the upper Bollinger Band.

Custom Strategy enters when EMA20 is above SMA50, CMF is positive, OBV is rising, and RSI is between 50 and 70.

Custom Strategy exits when EMA20 drops below SMA50, CMF turns negative, or RSI falls below 45.

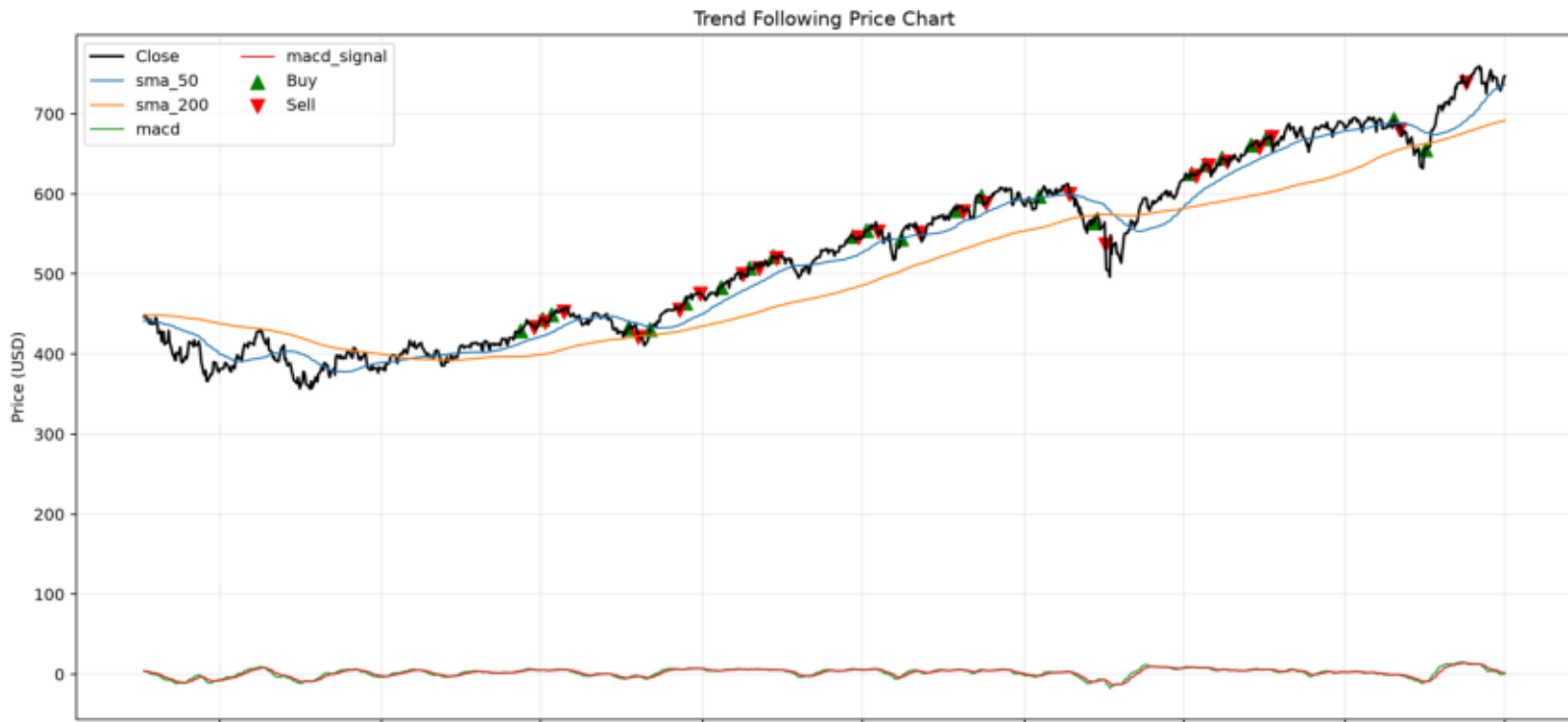
Discussion of results

Use the table and charts in this report to compare total return, risk-adjusted performance, drawdowns, and trade behavior against buy-and-hold.

Performance Comparison Table

Strategy	Total Return	CAGR	Volatility	Sharpe Ratio	Sortino Ratio	Maximum Drawdown	Win Rate	Trades Executed
Buy & Hold	67.29%	12.99%	17.40%	0.79	1.07	-20.55%	N/A	0
Trend Following	11.67%	2.65%	5.64%	0.49	0.28	-11.63%	39.13%	23
Mean Reversion	36.37%	7.64%	9.52%	0.82	0.40	-13.72%	100.00%	4
Custom Strategy	24.03%	5.24%	8.97%	0.61	0.59	-11.60%	41.94%	31

Trend Following Price Chart



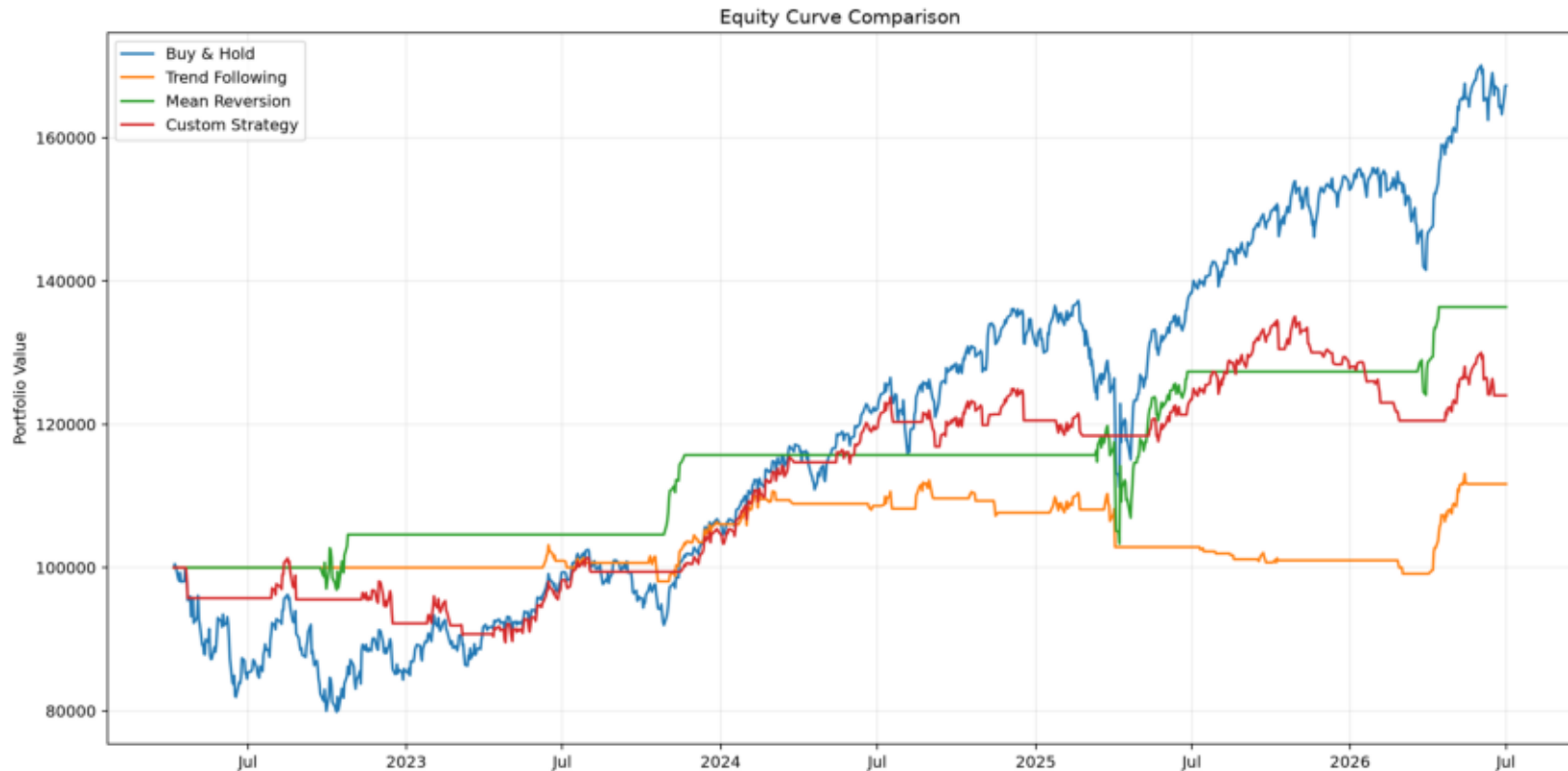
Mean Reversion Price Chart



Custom Strategy Price Chart



Equity Curve Comparison



Drawdown Comparison

